Physics of the Earth and Planetary Interiors, 9 (1974) 45–50
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THE PRIMARY MAGNETIC REMANENCE OF A DOLERITE SILL FROM NORTHEAST SPITSBERGEN

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Accepted for publication April 2, 1974

An Upper Jurassic dolerite sill from Lomfjord, northeast Spitsbergen, has been submitted to rock- and palaeomagnetic studies. In the cooling stage the original titanomagnetite underwent a high-temperature alteration forming exsolution of ilmenite lamellas. Due to rapid cooling the border zones in part escaped the high-temperature alteration but instead low-temperature oxidation, producing at least a low-temperature metastable magnetic mineral, is fairly pronounced in these parts of the sill. Thermal demagnetization has established that both mineral phases carry the same general magnetization direction suggesting the total remanence (reverse) to be of deuteric origin. Consideration of the cooling rate indicates that the accumulated geomagnetic secular variation may cover a time span of the order of 10^3 years. The relative pole position is at 61°N, 210°E, a result which is in agreement with some other Mesozoic results from Spitsbergen, but in rather marked disagreement with other published data for the Mesozoic of Europe.

1. Introduction

Within the Svalbard archipelago outcrops of Mesozoic dolerites may be traced over an area of about 130,000 km^2; radiometric and stratigraphic age estimates (Gayer et al., 1966; Parker, 1966) have arrived at an age of 125–150 m.y. for these rocks.

Palaeopole determination on the Spitsbergen dolerites have earlier been attempted. Krumsiek et al. (1968) suggested a pole with coordinates 58°N, 178°E (but only samples from one of the five collected sites were accepted) while Spall's (1968) results indicated an anomalous palaeopole at 60°N, 2°E. The samples studied in these investigations were all from the Isfjorden area in central Spitsbergen, an area which underwent subsidence during the 'Western Spitsbergen Orogeny' in mid-Tertiary times (Harland, 1969). Recent palaeomagnetic work on the Isfjorden dolerites (Halvorsen, 1973) suggests a complex magnetic structure, probably caused by remagnetization, and reliable palaeomagnetic directions have not been obtained. It seems reasonable to assume, therefore, that the earlier controversial results are the consequence of a partial remagnetization in mid-Tertiary times.

The dolerite sill from Lomfjorden dealt with in this paper originates outside the zone of Tertiary orogenic activity, and it was therefore hoped that the original palaeomagnetic properties of the rock were better preserved. Also, Upper Jurassic palaeomagnetic data for Europe are highly needed because of the few results so far published (Svalbard is regarded as part of the European plate).

On the eastern side of Lomfjorden, a system of two parallel dolerite sills can be traced over a distance of about 20 km. This observation together with the uniform thickness throughout indicate that the dolerites were intruded at a rather great depth (Tyrrell and Sandford, 1933). The stratigraphic position of the rock suggests a depth of intrusion of the order of 1–2 km (Orvin, 1940).

The dolerites are grey, medium to fine-grained granular rocks in which shining cleavage faces of plagioclase laths stand out prominantly against a dark grey background of pyroxenes and iron ores. In thin section the rocks are seen to be built of a plexus of laths of plagioclase in ophitic relations with

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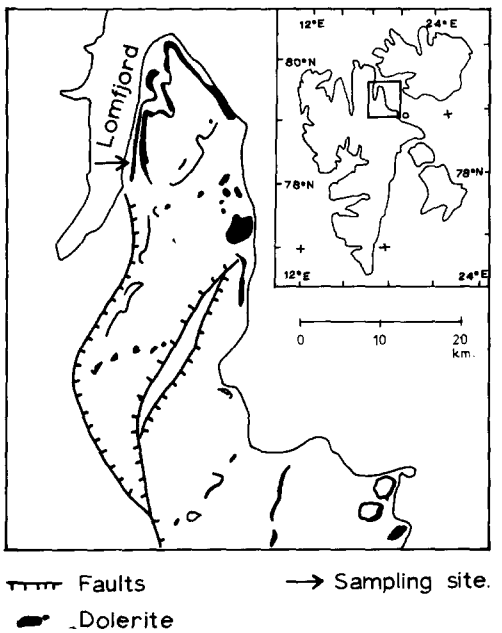


Fig. 1. Geological sketch map (after Orvin, 1940) showing the sampling area.

pyroxenes, interspersed with skeletal crystals of titaniferous iron ore, and involving patches of imperfectly crystallized mesostasial matter (Tyrrell and Sandford, 1933). A few irregular areas of red, non-pleochroic serpentine enclosed in the pyroxenes attest the former presence of olivine.

The lowermost Lomfjord sill, which penetrates the Cyathophyllum Limestone (Carboniferous), is 30–40 m thick. At the collecting site the sill is split into parts by a block of the country-rock. In order to study the magnetic properties of the sill, 38 hand samples were systematically collected from the lower margin up to the block. The Lomfjorden sill and the position of the sampled section are shown in the geological sketch map of Fig. 1. A more detailed sketch map which shows the sill and the country-rock with the vertical profile is given in Fig. 2.

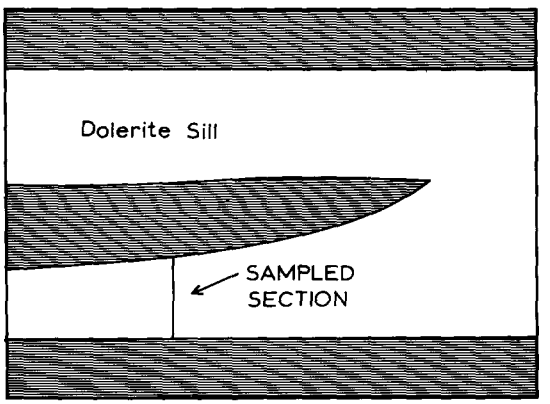


Fig. 2. A sketch map showing the position of the sampled section in the sill.

2. The magnetic minerals

2.1. Ore microscopy

Polished sections of samples from different levels in the sill have been studied under the ore microscope. The main ore minerals are ilmenite and titanomagnetite (T.M.) with accessory amounts of pyrite. The grain size increases from less than 10μ in the contact zone to more than 500μ in the middle of the sill.

All T.M. grains have exsolution lamellas of ilmenite indicating that the original T.M. has undergone a high temperature oxidation (Buddington and Lindsley, 1964).

The area between the lamellas is inhomogeneous and at least the following mineral phase or textures occur:

(1) Medium grey homogeneous areas suggesting Ti-poor titanomagnetite.

(2) Brownish areas, probably remainders of the primary Ti-rich T.M. or its low-temperature oxidation product (cation deficient spinel).

(3) Inhomogeneous black and grey areas, thought to be the commencement of a low-temperature alteration of Ti-rich T.M. called granulation (Ade-Hall et al., 1968).

The separate ilmenite grains are homogeneous, while the ilmenite lamellas are slightly altered, probably by exsolution of a magnesium-calcium spinel (Wilson and Haggerty, 1966). In the border zones of the sill, titanomagnetite and ilmenite are intergrown. This intergrowth is probably a crystallization phenomenon. Along the contacts the T.M. grains can be classified as oxidation-class 2 (Ade-Hall et al., 1968), but they often show the development of titanomaghemite phases. In the central part of the sill the T.M. belongs to oxidation class 4–5.

The average oxide content is about 6 vol. % through-

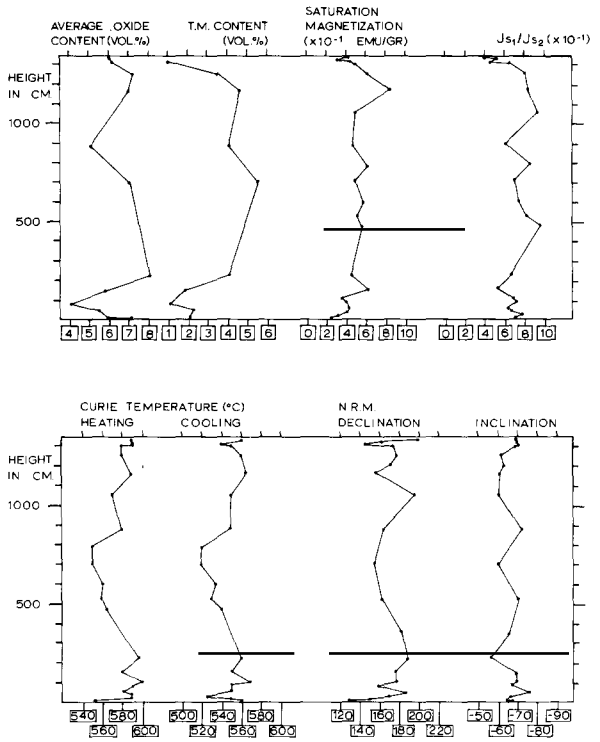


Fig. 3. The upper diagram shows the variation of the average oxide content, the titanomagnetite (T.M.) content, saturation magnetization (J_S) and the reduction of J_S after heating (J_{S1}/J_{S2}) in the sampled section. Curie temperatures before and after heating and NRM directions are plotted in the lower diagram.

out the traverse, but with a characteristic difference in magnetic mineralogy between the central part and the border zones, i.e., the T.M. content shows a maximum in the middle of the sill where the ilmenite is at a minimum. The reason for this difference is probably found in the thermal history of the rock. The variation in the average oxide content and the T.M. content throughout the traverse is shown in Fig. 3.

2.2. J_S–T measurements

The saturation magnetization (J_S) as a function of temperature has been measured with a translation balance. The heating was carried out in air with applied field strengths varying between 3000 Oe and 5000 Oe. The results can be divided into two groups as follows:

(1) A single-phase type with an apparent Curie point between 550°C and 600°C. After heating to

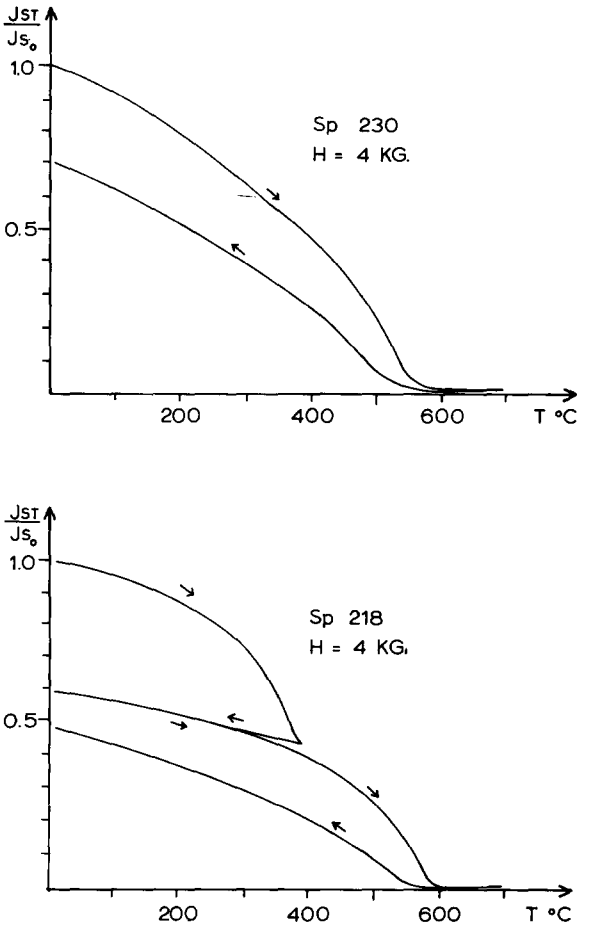


Fig. 4. Typical thermomagnetic curves from the border (SP 218) and from the middle (SP 230) of the sill.

700°C the T_c became reduced by 20–50°C while the J_S at room temperature was 0.5–0.9 times the original value (Fig. 4).

(2) Samples with two phases, T_c's at 350–400°C and at 550–600°C, respectively. The low-temperature phase disappears after heating to 700°C (Fig. 4, sample no. SP 218). Both T_c and J_S of the high-temperature phase show a similar reduction after heating to 700°C as the single-phase samples.

The double-phase samples are all found at the border zones of the sill while the single-phase samples are from the centre. It is assumed that the 550–600°C phase represents a Ti-poor spinel, which is consistent with the ore microscopy examination.

In order to examine in more detail the reversibility

of the 400°C-phase, samples were heated to 400°C, cooled down to room temperatures, and then heated to 700°C. As shown in Fig. 4 the 400°C-inflection is irreversible. Thus, this inflection can more likely be interpreted as a breakdown temperature rather than a true Curie point. The 400°C-phase involved is probably a cation deficient spinel breaking down to another mineral assemblage (Readman and O'Reilly, 1970) in which the magnetic component has a lower specific magnetization. J_s was measured down to liquid nitrogen temperature, but no magnetic phases below room temperature were discovered. A summary of the results obtained in the magnetomineralogical study is given in Fig. 3. J_{s1} and J_{s2} is the saturation magnetization before and after heating, respectively.

3. NRM-measurements and demagnetization results

The remanence directions of all 152 cylinders were found to be rather well grouped with a high negative inclination. The average remanent magnetic intensity for untreated specimens was $4.4 \cdot 10^{-3}$ emu/cm^3, with 99% of the measurements within the range of 10^{-3} emu/cm^3.

3.1. AC-demagnetization

Twenty cylinders, evenly spaced throughout the investigated traverse, were demagnetized. The results from all cylinders confirmed a stable direction with a high negative inclination. After demagnetization to more than 790 Oe only two cylinders showed directional changes. The majority of the samples kept their intensity of remanence (J_n) relatively unchanged when demagnetized to 200 Oe. Typical demagnetization results are given in Fig. 5.

3.2. Thermal demagnetization

The thermal demagnetization revealed the same stability of directions as the AC-treatment. Almost all the samples measured are without directional changes. Also, the stable directions from different samples are well grouped.

All the samples had a range of blocking temperatures between 200 and 550°C. Specimens from the lower and upper part of the investigated traverse

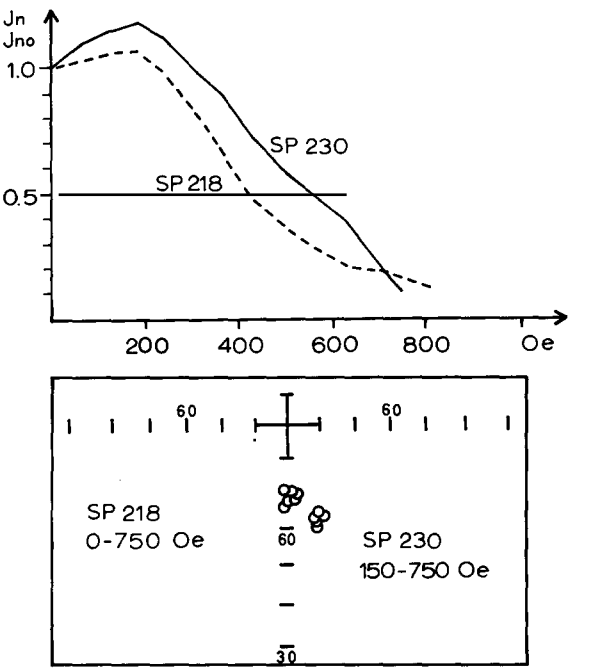


Fig. 5. Remanent intensity and magnetic direction obtained during progressive AC-demagnetization, stereographic projection.

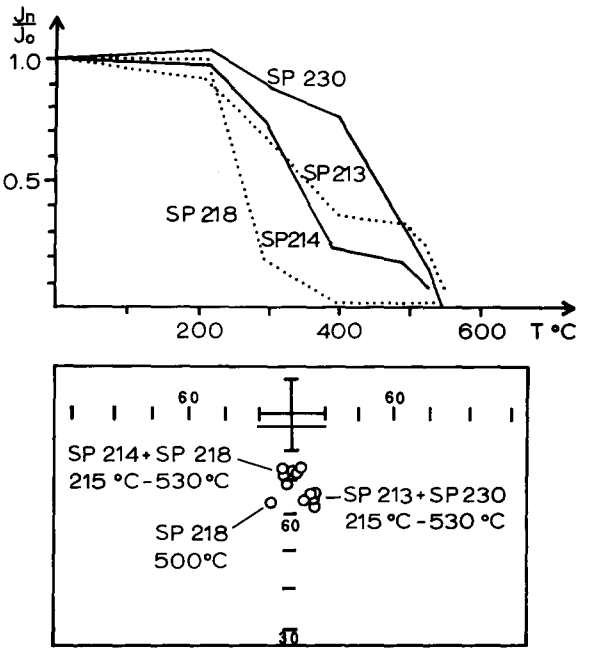


Fig. 6. Selected results from thermal-demagnetization experiments, stereographic projection.

revealed blocking temperatures mainly in the intervall 300–400°C while the blocking temperature spectrum was shifted to higher temperatures for specimens from the central part of the traverse. Typical demagnetization results are given in Fig. 6.

In the lower part of the sill the importance of the low blocking temperature phase decreases with increasing distance from the lower contact. Thus, in sample SP 218, which is sampled 1.5 m above the lower contact, only the low-temperature phase is found. The remanence decay at 300–400°C can be a real blocking temperature, but J_s–T measurements suggest inversion of a metastable mineral as a more likely explanation.

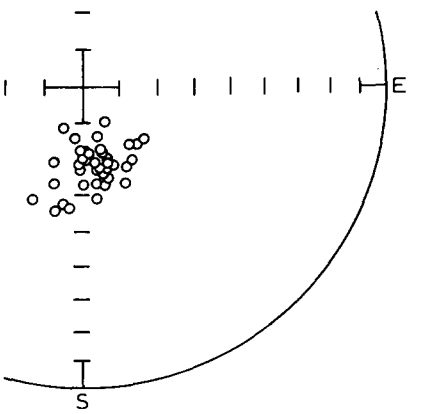


Fig. 7. Stable magnetization directions from 38 samples, equal-area projection.

4. Discussion

The magnetomineralogical study of the investigated rocks has revealed a significant difference in magnetic mineralogy between samples from the border zones and those from the central part of the section. Study of the saturation magnetization versus temperature showed a two-phase magnetic mineralogy in the border zones but only single-phase observations emerged in the central part of the traverse. The two magnetic phases involved is firstly a high-temperature one (T_c = 550°C) believed to be Ti-poor titanomagnetite formed through exsolution–oxidation processes during the original cooling and secondly a low-temperature phase thought to have originated from a subsequent low-temperature oxidation. The latter phase is found to invert during heating; any magnetic product resulting from the inversion has a too low specific magnetization to be recognized in the J_s–T experiments. The Ti-poor spinel, believed to constitute the high-temperature phase, shows a reduction in T_c of about 20–50°C after heating, a feature open to several interpretations none of which seems clearly indicated in the present case. The variation of bulk magnetic mineralogy is also demonstrated in the remanence properties. On thermal demagnetization the intensity decay curves reveal reduction in two main temperature intervals: 300–400°C and 500–550°C. Thus, it looks as if both phases contribute to the NRM.

The low-temperature phase which seems to be responsible for a substantial part of the NRM might have been formed during the original cooling of the rock. According to Jaeger (1968) the investigated sill, which seems to have been emplaced at a depth of 1–2 km, probably needed a time of the order of 10^3 years to cool down to temperatures 50°C above the surroundings. Due to the extensive Mesozoic igneous activity in the area of Lomfjorden, it is reasonable to assume a quite large geothermal gradient for the region and a temperature level of 100–200°C at the depth of intrusion seems quite likely. The fact that the magnetization directions of the two phases are identical points to a deuteric origin of the low-temperature magnetic constituent. Another possibility is that there was no apparent polar wandering or geomagnetic polarity change in the period from the formation of the high-temperature magnetization (believed to be primary) to the formation of low-temperature magnetization which in this case could be a post-deuteric process. Due to the suggested amount of continental drift in the Late Jurassic/Early Cretaceous (Pitman III and Talwani, 1972) the latter explanation is considered less likely than the first one.

The assumed slow cooling rate of the sill probably implies that the build-up of the remanence conforms more to the structure of a moderate temperature VRM (Storetvedt, 1968) than to that of a TRM in its strict sense. One must assume, therefore, that a major part of geomagnetic secular variation has been averaged out; the mean direction, therefore, serving as

a reasonable basis for a palaeopole estimation. The stable magnetization direction from all 38 samples (Fig. 7) gives the following results: $D = 171°, I = -68°$, $\alpha_{95} = 3°, K = 95.9$ (unit weight on sample) and the estimated pole has the coordinates of 61°N, 210°E. Though there are obvious limitations in the geophysical significance of palaeomagnetic data of a small rock body it is interesting to realize that the present pole estimate exhibits the same easterly position as the Late Cretaceous palaeopole for Svalbard recently presented by Sandal and Halvorsen (1973). These positions are quite different from other Jurassic and Late Cretaceous poles for Europe as for instance those tabulated by Phillips and Forsyth (1972). Further work in progress on dolerites from the Hinlopen area of Spitsbergen seems to line up with the result here presented. This probably implies that the Mesozoic polar path relative to Europe has to be modified.

Acknowledgment

We are grateful to Professor K.M. Storetvedt for discussion and critical evaluation of the manuscript. Norsk Polarinstitutt is thanked for extensive assistance during the field work. Financial assistance for the Spitsbergen project has been provided by the Norwegian Research Council for Science and the Humanities.

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